

--TASK 1 NESTED LOOP JOIN

Nested Loop is the only option for inequality condition for such a query. A hash and merge require search on equal conditions

```
EXPLAIN ANALYZE
SELECT * FROM test_joins_a a, test_joins_b b
WHERE a.id1 > b.id1;
```

1	×	Results 2
ANALYZE SELECT * FROM test_joins_a a, test_joins_b b WHERE a.id1 > b.id1; Enter a SQL expression to filter results (use Ctrl+Space)		
A-Z QUERY PLAN		
Nested Loop (cost=0.00..1500315.00 rows=33333333 width=16) (actual time=2.641..17748.005 rows=49995000.00 loops=1)		
Join Filter: (a.id1 > b.id1)		
Rows Removed by Join Filter: 50005000		
Buffers: shared hit=90		
-> Seq Scan on test_joins_a a (cost=0.00..145.00 rows=10000 width=8) (actual time=0.018..3.246 rows=10000.00 loops=1)		
Buffers: shared hit=45		
-> Materialize (cost=0.00..195.00 rows=10000 width=8) (actual time=0.000..0.481 rows=10000.00 loops=10000)		
Storage: Memory Maximum Storage: 441kB		
Buffers: shared hit=45		
-> Seq Scan on test_joins_b b (cost=0.00..145.00 rows=10000 width=8) (actual time=0.008..0.544 rows=10000.00 loops=1)		
Buffers: shared hit=45		
Planning:		
Buffers: shared hit=20		
Planning Time: 1.462 ms		
Execution Time: 19730.487 ms		

The second is an explicit cross join. Every row of A pairs with every row of B, so there's no join condition at all to hash or sort on. Nested Loop is structurally the only way to produce full combination.

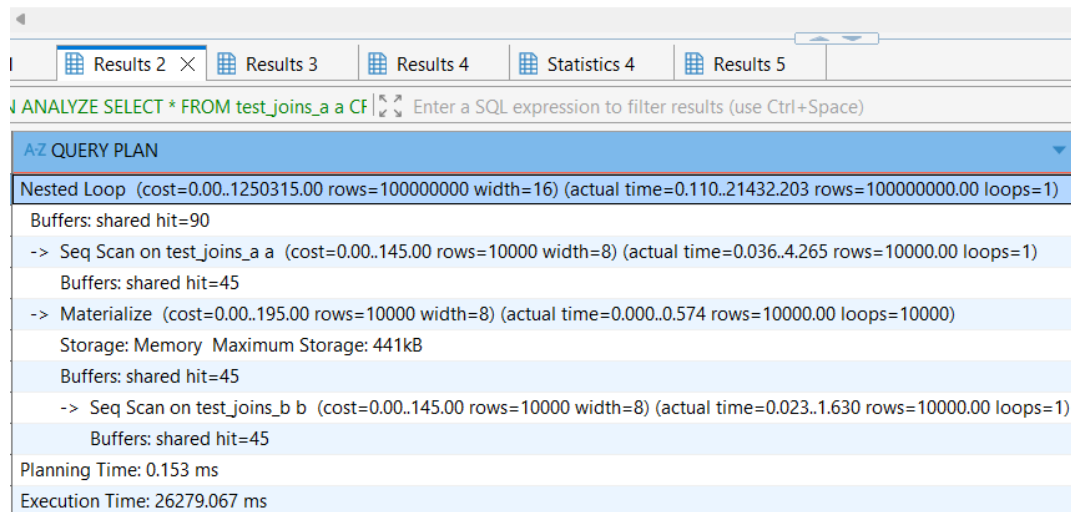
```
EXPLAIN ANALYZE
SELECT *
FROM test_joins_a a
CROSS JOIN test_joins_b b;
```

1	×	Results 2
ANALYZE SELECT * FROM test_joins_a a CROSS JOIN test_joins_b b; Enter a SQL expression to filter results (use Ctrl+Space)		
A-Z QUERY PLAN		
Nested Loop (cost=0.00..1250315.00 rows=100000000 width=16) (actual time=0.110..21432.203 rows=100000000.00 loops=1)		
Buffers: shared hit=90		
-> Seq Scan on test_joins_a a (cost=0.00..145.00 rows=10000 width=8) (actual time=0.036..4.265 rows=10000.00 loops=1)		
Buffers: shared hit=45		
-> Materialize (cost=0.00..195.00 rows=10000 width=8) (actual time=0.000..0.574 rows=10000.00 loops=10000)		
Storage: Memory Maximum Storage: 441kB		
Buffers: shared hit=45		
-> Seq Scan on test_joins_b b (cost=0.00..145.00 rows=10000 width=8) (actual time=0.023..1.630 rows=10000.00 loops=1)		
Buffers: shared hit=45		
Planning Time: 0.153 ms		
Execution Time: 26279.067 ms		

--TASK 2 HASH JOIN

The hash join is applied after the > is changed to = . One of the tables is hashed to the memory (Buckets: 16384 Batches: 1 Memory Usage: 519kB). This is faster than nested loops because the one table scanned through the hash table once.

```
EXPLAIN ANALYZE
SELECT * FROM test_joins_a a, test_joins_b b
WHERE a.id1 = b.id1;
```



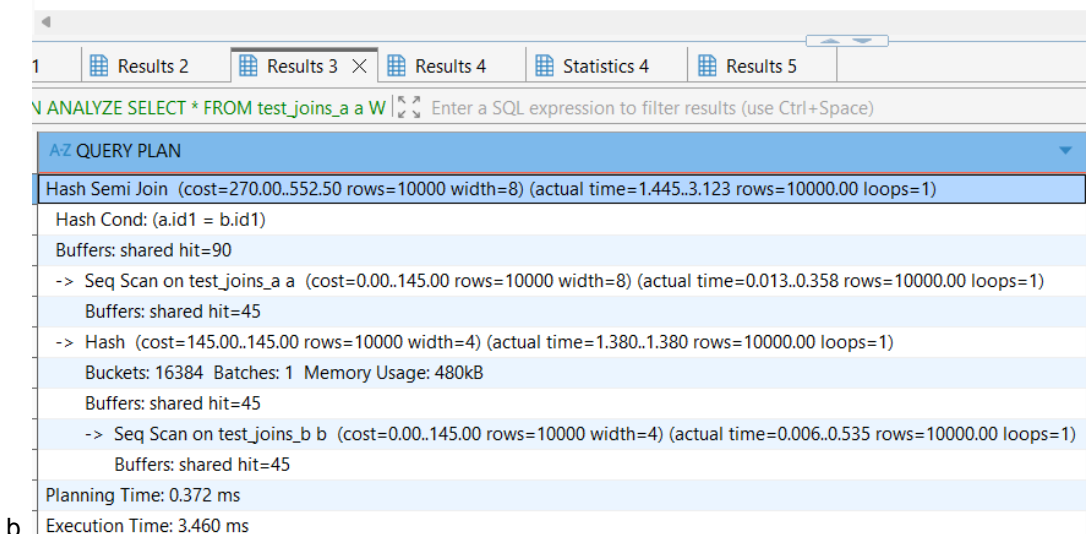
The screenshot shows the query plan for the SQL query: `SELECT * FROM test_joins_a a, test_joins_b b WHERE a.id1 = b.id1;`. The plan is a Nested Loop join. It first performs a sequential scan on `test_joins_a` (10,000 rows, 8 bytes width), then materializes it into memory (441kB storage, 10,000 rows). For each row in the materialized table, it performs a sequential scan on `test_joins_b` (10,000 rows, 8 bytes width). The total cost is 0.00..1250315.00, and the actual execution time is 0.110..21432.203 ms. Buffers: shared hit=90.

Step	Operation	Cost	Rows	Width	Actual Time	Actual Rows	Actual Width	Loops
1	Nested Loop	0.00..1250315.00	100000000	16	0.110..21432.203	100000000	16	1
2	Seq Scan on test_joins_a a	0.00..145.00	10000	8	0.036..4.265	10000	8	1
3	Materialize	0.00..195.00	10000	8	0.000..0.574	10000	8	10000
4	Seq Scan on test_joins_b b	0.00..145.00	10000	8	0.023..1.630	10000	8	1

Planning Time: 0.153 ms
Execution Time: 26279.067 ms

WHERE EXISTS map to semi joins, which return each row of the left table once regardless of how many matches exist on the right side. Hash Join makes sense here since each id1 in a has exactly one match in

```
EXPLAIN ANALYZE
SELECT * FROM test_joins_a a
WHERE EXISTS (
  SELECT 1 FROM test_joins_b b WHERE b.id1 = a.id1
);
```



The screenshot shows the query plan for the SQL query: `SELECT * FROM test_joins_a a WHERE EXISTS (SELECT 1 FROM test_joins_b b WHERE b.id1 = a.id1);`. The plan is a Hash Semi Join. It first performs a sequential scan on `test_joins_a` (10,000 rows, 8 bytes width). Then it hashes the `id1` values (16384 buckets, 1 batch, 519kB memory usage). For each row in the scanned table, it checks the hash table (10,000 rows, 4 bytes width). The total cost is 270.00..552.50, and the actual execution time is 1.445..3.123 ms. Buffers: shared hit=90.

Step	Operation	Cost	Rows	Width	Actual Time	Actual Rows	Actual Width	Loops
1	Hash Semi Join	270.00..552.50	10000	8	1.445..3.123	10000	8	1
2	Seq Scan on test_joins_a a	0.00..145.00	10000	8	0.013..0.358	10000	8	1
3	Hash	145.00..145.00	10000	4	1.380..1.380	10000	4	1
4	Seq Scan on test_joins_b b	0.00..145.00	10000	4	0.006..0.535	10000	4	1

Planning Time: 0.372 ms
Execution Time: 3.460 ms

Turning off `enable_hashjoin` shifts the query to Nested Loop that process the query, but taking 6885ms instead of 3 ms, significantly slower, because now it has to check all combinations and filter down instead of doing a direct hash lookup.

```
SET enable_hashjoin = off;  
EXPLAIN ANALYZE  
SELECT * FROM test_joins_a a, test_joins_b b  
WHERE a.id1 = b.id1;
```

1	Results 2	Results 3	Results 4	Statistics 4	Results 5
ble_hashjoin = off; EXPLAIN ANALYZE SELECT * FROM test_joins_a a, test_joins_b b WHERE a.id1 = b.id1; Enter a SQL expression to filter results (use Ctrl+Space)					
A-Z QUERY PLAN					
Nested Loop (cost=0.00..1500315.00 rows=10000 width=16) (actual time=0.031..6883.742 rows=10000.00 loops=1)					
Disabled: true					
Join Filter: (a.id1 = b.id1)					
Rows Removed by Join Filter: 99990000					
Buffers: shared hit=90					
-> Seq Scan on test_joins_a a (cost=0.00..145.00 rows=10000 width=8) (actual time=0.011..1.453 rows=10000.00 loops=1)					
Buffers: shared hit=45					
-> Materialize (cost=0.00..195.00 rows=10000 width=8) (actual time=0.000..0.245 rows=10000.00 loops=10000)					
Storage: Memory Maximum Storage: 441kB					
Buffers: shared hit=45					
-> Seq Scan on test_joins_b b (cost=0.00..145.00 rows=10000 width=8) (actual time=0.006..0.347 rows=10000.00 loops=1)					
Buffers: shared hit=45					
Planning Time: 0.061 ms					
Execution Time: 6884.494 ms					

Turning on `enable_hashjoin` back shows quick results(3ms)

```
SET enable_hashjoin = on;  
EXPLAIN ANALYZE  
SELECT * FROM test_joins_a a, test_joins_b b  
WHERE a.id1 = b.id1;
```

1	Results 2	Results 3	Results 4	Statistics 4	Results 5
able_hashjoin = on; EXPLAIN ANALYZE SELECT * FROM test_joins_a a, test_joins_b b WHERE a.id1 = b.id1; Enter a SQL expression to filter results (use Ctrl+Space)					
A-Z QUERY PLAN					
Hash Join (cost=270.00..552.50 rows=10000 width=16) (actual time=1.201..3.108 rows=10000.00 loops=1)					
Hash Cond: (a.id1 = b.id1)					
Buffers: shared hit=90					
-> Seq Scan on test_joins_a a (cost=0.00..145.00 rows=10000 width=8) (actual time=0.018..0.328 rows=10000.00 loops=1)					
Buffers: shared hit=45					
-> Hash (cost=145.00..145.00 rows=10000 width=8) (actual time=1.128..1.129 rows=10000.00 loops=1)					
Buckets: 16384 Batches: 1 Memory Usage: 519kB					
Buffers: shared hit=45					
-> Seq Scan on test_joins_b b (cost=0.00..145.00 rows=10000 width=8) (actual time=0.005..0.355 rows=10000.00 loops=1)					
Buffers: shared hit=45					
Planning Time: 0.064 ms					
Execution Time: 3.411 ms					

--TASK 3 MERGE JOIN

The Hash Join is still preferable in equal conditions, merge is preferable when hash is off only, the postgres needs extra time to sort tables.

```
SET enable_mergejoin = off;  
EXPLAIN ANALYZE  
SELECT * FROM test_joins_a a, test_joins_b b  
WHERE a.id1 = b.id1;
```

Results 3 Results 4 Results 5 Statistics 6 Statistics 7 Results 8 Results

mergejoin = off; EXPLAIN ANALYZE SELECT * FROM test_joins_a a, test_joins_b b WHERE a.a_id1 = b.b_id1;

Enter a SQL expression to filter results (use Ctrl+Space)

AZ QUERY PLAN

Hash Join (cost=270.00..552.50 rows=10000 width=16) (actual time=1.397..4.036 rows=10000.00 loops=1)

Hash Cond: (a.id1 = b.id1)

Buffers: shared hit=90

-> Seq Scan on test_joins_a a (cost=0.00..145.00 rows=10000 width=8) (actual time=0.019..0.448 rows=10000.00 loops=1)

Buffers: shared hit=45

-> Hash (cost=145.00..145.00 rows=10000 width=8) (actual time=1.310..1.311 rows=10000.00 loops=1)

Buckets: 16384 Batches: 1 Memory Usage: 519kB

Buffers: shared hit=45

-> Seq Scan on test_joins_b b (cost=0.00..145.00 rows=10000 width=8) (actual time=0.008..0.417 rows=10000.00 loops=1)

Buffers: shared hit=45

Planning Time: 0.103 ms

Execution Time: 4.420 ms

```
SET enable_mergejoin = on;
EXPLAIN ANALYZE
SELECT * FROM test_joins_a a, test_joins_b b
WHERE a.id1 = b.id1;
```

	Statistics 6	Statistics 7	Results 8	Results 9	Results 10	Statistics 10	Re
ble_mergejoin = on; EXPLAIN ANALYZE SEL Enter a SQL expression to filter results (use Ctrl+Space)							
A-Z QUERY PLAN							
Merge Join (cost=1618.77..1818.77 rows=10000 width=16) (actual time=1.837..4.182 rows=10000.00 loops=1)							
Merge Cond: (a.id1 = b.id1)							
Buffers: shared hit=90							
-> Sort (cost=809.39..834.39 rows=10000 width=8) (actual time=0.904..1.206 rows=10000.00 loops=1)							
Sort Key: a.id1							
Sort Method: quicksort Memory: 619kB							
Buffers: shared hit=45							
-> Seq Scan on test_joins_a a (cost=0.00..145.00 rows=10000 width=8) (actual time=0.013..0.357 rows=10000.00 loops=1)							
Buffers: shared hit=45							
-> Sort (cost=809.39..834.39 rows=10000 width=8) (actual time=0.929..1.167 rows=10000.00 loops=1)							
Sort Key: b.id1							
Sort Method: quicksort Memory: 619kB							
Buffers: shared hit=45							
-> Seq Scan on test_joins_b b (cost=0.00..145.00 rows=10000 width=8) (actual time=0.008..0.302 rows=10000.00 loops=1)							
Buffers: shared hit=45							
Planning Time: 0.101 ms							
Execution Time: 4.549 ms							

-- TASK 4 CHANGING JOIN ORDER

Forcing join_collapse_limit = 1 reflects the literal order: b JOIN a first, then joined to c.(hash right join) With the default join_collapse_limit = 8, Postgres reordered the joins differing from the literal query order (b JOIN a, then LEFT JOIN c)(hash join)

```
SET join_collapse_limit = 1;
```

EXPLAIN

```
SELECT c.id2
FROM test_joins_b b
JOIN test_joins_a a ON (b.id1 = a.id1)
LEFT JOIN test_joins_c c ON (c.id1 = b.id1);
```

1	Results 2	Results 4	Results 4	Statistics 5	
SELECT c.id2 FROM test_joins_b b JOIN tes Enter a SQL expression to filter results					
AZ QUERY PLAN					
Hash Right Join (cost=677.50..18952.50 rows=10000 width=4)					
Hash Cond: (c.id1 = b.id1)					
-> Seq Scan on test_joins_c c (cost=0.00..14425.00 rows=1000000 width=8)					
-> Hash (cost=552.50..552.50 rows=10000 width=4)					
-> Hash Join (cost=270.00..552.50 rows=10000 width=4)					
Hash Cond: (b.id1 = a.id1)					
-> Seq Scan on test_joins_b b (cost=0.00..145.00 rows=10000 width=4)					
-> Hash (cost=145.00..145.00 rows=10000 width=4)					
-> Seq Scan on test_joins_a a (cost=0.00..145.00 rows=10000 width=4)					

EXPLAIN

```
SELECT c.id2
FROM test_joins_b b
JOIN test_joins_a a on (b.id1 = a.id1)
LEFT JOIN test_joins_c c on (c.id1 = b.id1);
```

1	Results 2	Results 4	Results 4	Statistics 5	
SELECT c.id2 FROM test_joins_b b JOIN tes Enter a SQL expression to filter results					
AZ QUERY PLAN					
Hash Join (cost=540.00..18952.50 rows=10000 width=4)					
Hash Cond: (b.id1 = a.id1)					
-> Hash Right Join (cost=270.00..18545.00 rows=10000 width=8)					
Hash Cond: (c.id1 = b.id1)					
-> Seq Scan on test_joins_c c (cost=0.00..14425.00 rows=1000000 width=8)					
-> Hash (cost=145.00..145.00 rows=10000 width=4)					
-> Seq Scan on test_joins_b b (cost=0.00..145.00 rows=10000 width=4)					
-> Hash (cost=145.00..145.00 rows=10000 width=4)					
-> Seq Scan on test_joins_a a (cost=0.00..145.00 rows=10000 width=4)					

--TASK 5 LATERAL JOIN

```
SELECT
  s.store_id,
  s.store_name,
  o.order_id,
  o.order_num,
  o.order_cost
FROM stores s
CROSS JOIN LATERAL (
  SELECT
    order_id,
    order_num,
    order_cost
  FROM orders
  WHERE order_cost <= s.max_order_cost
  ORDER BY order_cost DESC
  LIMIT 10
) o
ORDER BY
  s.store_id,
  o.order_cost DESC;
```

123 store_id	AZ store_name	123 order_id	AZ order_num	123 order_cost
1	grossery shop	11	order number 11	776
1	grossery shop	20	order number 20	763
1	grossery shop	379	order number 379	745
1	grossery shop	15	order number 15	744
1	grossery shop	950	order number 950	729
1	grossery shop	180	order number 180	729
1	grossery shop	139	order number 139	717
1	grossery shop	106	order number 106	701
1	grossery shop	18	order number 18	700
1	grossery shop	428	order number 428	676
2	bakery	3	order number 3	71
2	bakery	1	order number 1	50
2	bakery	2	order number 2	42
2	bakery	21	order number 21	41
2	bakery	4	order number 4	3
3	manufactured goods	38	order number 38	2,961
3	manufactured goods	31	order number 31	2,960
3	manufactured goods	60	order number 60	2,955
3	manufactured goods	152	order number 152	2,849
3	manufactured goods	67	order number 67	2,843
3	manufactured goods	346	order number 346	2,763
3	manufactured goods	70	order number 70	2,750
3	manufactured goods	372	order number 372	2,679
3	manufactured goods	393	order number 393	2,656
3	manufactured goods	888	order number 888	2,655

Due to lateral join type, the FROM subquery runs once per row of stores, each time substituting that store's own max_order_cost, and returns up to 10 matching orders per store. This resembles for each loop approach and allows to scan table for required values.

--TASK 6 RECURSIVE CTE

```
WITH RECURSIVE emp_hierarchy AS (  
    SELECT empno, mgr, ename, ename AS mgrname, 1 AS lvl  
    FROM emp  
    WHERE mgr IS NULL  
    UNION ALL  
    SELECT e.empno, e.mgr, e.ename, h.ename AS mgrname, h.lvl + 1 AS lvl  
    FROM emp e  
    JOIN emp_hierarchy h ON e.mgr = h.empno  
)  
SELECT empno, mgr, ename, mgrname, lvl  
FROM emp_hierarchy  
ORDER BY lvl, empno;
```

empno	mgr	ename	mgrname	lvl
9	[NULL]	KING	KING	1
4	9	JONES	KING	2
6	9	BLAKE	KING	2
7	9	CLARK	KING	2
2	6	ALLEN	BLAKE	3
3	6	WARD	BLAKE	3
5	6	MARTIN	BLAKE	3
8	4	SCOTT	JONES	3
10	6	TURNER	BLAKE	3
12	6	JAMES	BLAKE	3
13	4	FORD	JONES	3
1	13	SMITH	FORD	4
11	8	ADAMS	SCOTT	4
14	1	KELLY	SMITH	5
15	1	HANNAH	SMITH	5

Firstly, the president (mgr IS NULL) was selected with assigning lvl = 1.

Next, joins emp to the emp_hierarchy - find employees whose manager is someone we already placed in the hierarchy.

Query keeps re-running against the newly added rows from the previous pass, stopping automatically once a pass returns zero new rows.

--TASK 7 CHANGING DATA CTE

```
WITH updated_rows AS (  
    UPDATE orders  
    SET order_cost = order_cost / 2  
    WHERE order_cost BETWEEN 100 AND 1000  
    RETURNING order_id, order_cost, order_num  
)  
,  
deleted_rows AS (  
    DELETE FROM orders  
    WHERE order_cost < 50  
    RETURNING order_id, order_cost, order_num  
)  
INSERT INTO order_log (order_id, order_cost, order_num, action_type)  
SELECT order_id, order_cost, order_num, 'U' FROM updated_rows  
UNION ALL  
SELECT order_id, order_cost, order_num, 'D' FROM deleted_rows;
```

Results 1	Results 2	Results 4	Results 4	Statistics 5	order_log 6	orders 9
SELECT * FROM order_log ORDER BY log_date						
log_id	order_id	order_cost	order_num	action_type	log_date	
39	451	209	order number 451	U	2026-07-14 22:25:15.973 +0300	
40	503	301	order number 503	U	2026-07-14 22:25:15.973 +0300	
41	561	70	order number 561	U	2026-07-14 22:25:15.973 +0300	
42	653	270	order number 653	U	2026-07-14 22:25:15.973 +0300	
43	712	405	order number 712	U	2026-07-14 22:25:15.973 +0300	
44	941	119	order number 941	U	2026-07-14 22:25:15.973 +0300	
45	950	364	order number 950	U	2026-07-14 22:25:15.973 +0300	
46	979	488	order number 979	U	2026-07-14 22:25:15.973 +0300	
47	2	42	order number 2	D	2026-07-14 22:25:15.973 +0300	
48	4	3	order number 4	D	2026-07-14 22:25:15.973 +0300	
49	21	41	order number 21	D	2026-07-14 22:25:15.973 +0300	

The WITH statement combines an UPDATE (change cost for rows 100–1000) and a DELETE (remove rows with cost < 50), both with RETURNING, adding into an INSERT that logs both actions to order_log with action type 'U'/'D'.